1. Customer Churn Prediction

- **Problem Statement**: A telecom company wants to predict which customers are likely to churn in the next month.
- **Objective**: Develop a predictive model to identify customers at risk of leaving, allowing the company to take preventive actions.
- **Constraints**:
    - Imbalanced dataset (churned vs. non-churned customers).
    - Real-time prediction required for proactive intervention.
    - Privacy concerns regarding customer data.

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2. Demand Forecasting for Retail

- **Problem Statement**: A retail chain wants to predict demand for different products to optimize inventory management.
- **Objective**: Build a demand forecasting model to reduce stockouts and overstock situations.
- **Constraints**:
    - Seasonal variations and market trends affect demand.
    - Availability of historical sales data might be limited.
    - Need for region-wise predictions for different stores.

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3. Fraud Detection in Banking

- **Problem Statement**: A bank wants to detect fraudulent credit card transactions in real-time.
- **Objective**: Develop a machine learning model to identify fraudulent transactions with high accuracy.
- **Constraints**:
    - Highly imbalanced dataset (fraud cases are rare).
    - Need for real-time inference to prevent fraud.
    - False positives can inconvenience genuine customers.

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4. Personalized Product Recommendations

- **Problem Statement**: An e-commerce platform wants to improve customer experience by recommending relevant products.
- **Objective**: Build a recommendation system that suggests products based on user behavior and preferences.
- **Constraints**:
    - Large-scale data processing for millions of users.
    - Cold start problem for new users with no history.
    - Real-time recommendations required for a seamless user experience.

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5. Predictive Maintenance in Manufacturing

- **Problem Statement**: A manufacturing company wants to predict machine failures before they occur.
- **Objective**: Develop a predictive maintenance model to reduce downtime and maintenance costs.
- **Constraints**:
    - Availability of sensor data and maintenance logs.
    - Need for interpretable models to understand failure causes.
    - High cost of false negatives (missed failures).

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6. Healthcare Disease Prediction

- **Problem Statement**: A hospital wants to predict the likelihood of patients developing diabetes based on their health records.
- **Objective**: Build a predictive model to assist doctors in early diagnosis and intervention.
- **Constraints**:
    - Need for high accuracy due to medical implications.
    - Data privacy and regulatory compliance (HIPAA, GDPR).
    - Limited and biased historical data.

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7. Sentiment Analysis for Brand Monitoring

- **Problem Statement**: A brand wants to analyze customer sentiment on social media to track its reputation.
- **Objective**: Build an NLP-based model to classify customer reviews and social media posts as positive, neutral, or negative.
- **Constraints**:
    - Large volume of unstructured text data.

- Need for language and slang interpretation.
- Sentiment analysis can be subjective and context-dependent.

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8. Loan Default Prediction

- **Problem Statement**: A financial institution wants to assess the risk of customers defaulting on their loans.
- **Objective**: Develop a credit risk model to improve loan approval decisions.
- **Constraints**:
  - Regulatory constraints on using sensitive customer data.
  - Highly imbalanced dataset (few defaulters).
  - Need for explainability to comply with financial regulations.

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9. Supply Chain Optimization

- **Problem Statement**: A logistics company wants to optimize delivery routes to reduce costs and improve efficiency.
- **Objective**: Build an optimization model to minimize delivery time and fuel consumption.
- **Constraints**:
  - Dynamic traffic conditions and weather impact routes.
  - Real-time decision-making required for rerouting.
  - Large-scale optimization problem with multiple constraints (e.g., vehicle capacity, delivery time windows).

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10. Employee Attrition Prediction

- **Problem Statement**: A company wants to predict which employees are likely to leave to improve retention strategies.
- **Objective**: Develop a predictive model to identify at-risk employees and take preventive measures.
- **Constraints**:
  - Sensitive HR data with privacy concerns.
  - Difficulty in quantifying employee dissatisfaction.
  - Need for explainable results for HR intervention.
